

Solar Home Lighting 3

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FUNCTIONAL SYSTEM REQUIREMENTS

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FUNCTIONAL SYSTEM REQUIREMENTS FOR Solar Home Lighting

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1. Introduction

1.1. Purpose and Scope

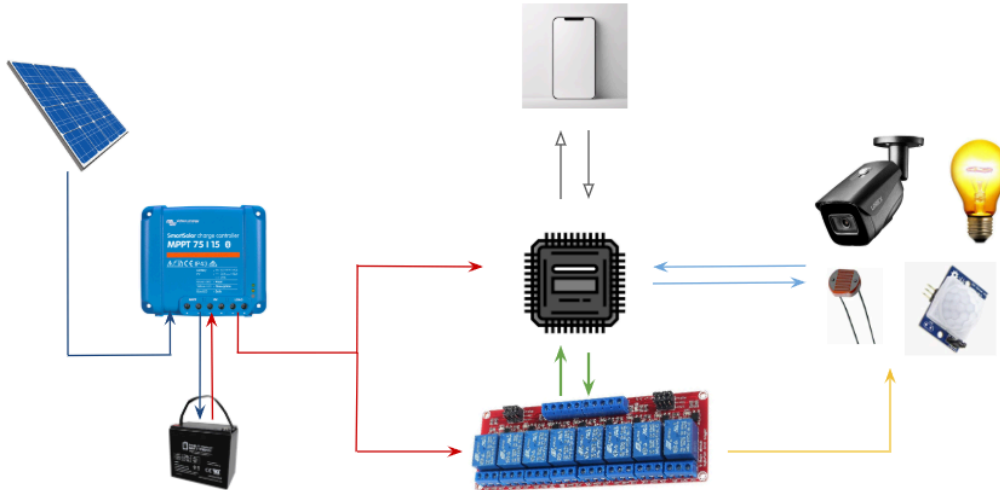
The purpose of this project is to design, develop, and demonstrate a fully integrated solar-powered residential lighting and security system capable of operating independently from the electrical grid. The system shall support both interior and exterior applications and shall combine the functionality of traditional lighting control with modern smart security features into a unified, renewable energy-powered platform.

This system encompasses the complete development of a modular system architecture, including solar energy generation, battery storage, sensor driven automation, real time monitoring, and user control via both a local interface and mobile application. The system is intended to enhance residential security while reducing energy consumption and utility costs by leveraging solar energy as the primary power source.

For demonstration purposes, a function prototype will be built, simulating a typical porch and foyer installation. However, the system is designed to be scalable and expandable to support multiple rooms and outdoor zones across a full residential property. The final implementation shall include motion detection, door sensors, environmental light sensing, programmable timers, and a network of lighting and surveillance devices. All of which will be managed through a central microcontroller and ESP32 based wireless communication.

The scope of this work includes hardware and software integration, embedded system development, sensor interfacing, application design, and functional validation of system performance under typical residential operating conditions.

Figure 1.1.1. Conceptual Product Layout



The Solar Home Lighting System uses sunlight to power lights and small devices in a home. During the day, the solar panel will collect energy from the sun and send it to a battery for storage via the charge controller. When the load is drawing power, power distribution will be controlled in accordance to user preference and power availability.

Users can connect to the system using a mobile app, which shows how much energy is being made, how full the battery is, and lets them turn lights on or off remotely. All the system's data, like how much power was used or if there were any problems, is saved in a database.

1.2. Responsibility and Change Authority

All project requirements shall be verified under the authority of the team captain, Jack Holdridge, who shall ensure each requirement is met in accordance with the project specification. No requirement may be altered without prior approval from both the team captain and the project sponsor, Professor Won Jang.

Subsystem Ownership is as follows:

- MPPT Charge Controller – Adam Garcia
- Physical Controls – Jack Holdridge
- Android Application and Camera – Kyle Walcher
- MCU/Sensors & Database – Matthew Greer

2. Applicable and Reference Documents

2.1. Applicable Documents

The following documents, of the exact issue and revision shown, form a part of this specification to the extent specified herein:

2.1.1. Applicable Documents Table

Document Number / Title	Revision / Release Date	Description
Lighting Global SHS Quality Standards v2.5	Dec 2018	<i>Defines minimum quality, durability, and safety for solar home system kits up to 350W. Includes testing protocols and eligibility criteria.</i>
IEEE 1547a	2020	<i>IEEE Standard for Interconnection and Interoperability of Distributed Energy Resources with Associated Electric Power Systems Interfaces--Amendment 1: To Provide More Flexibility for Adoption of Abnormal Operating Performance Category III</i>
NEC Article 690	2023	<i>National Electrical Code for Solar Photovoltaic Systems. Covers wiring, grounding, and safety for PV installations.</i>
UL 1741	2021	<i>Standard for Inverters, Converters, Controllers, and Interconnection System Equipment for Use With Distributed Energy Resources.</i>
ANSI C119.6	2024	<i>Electric Connectors—Non-Sealed, Multiport Connector Systems Rated 600 V or Less for Aluminum and Copper Conductors.</i>
IEC 62852	2014	<i>Connectors for DC-application in photovoltaic systems - Safety requirements and tests</i>
NEC Article 480	2023	<i>National Electrical Code for Storage Batteries</i>

2.2. Reference Documents

The following documents are reference documents utilized in the development of this specification. These documents do not form a part of this specification and are not controlled by their reference herein.

2.2.1. Reference Documents Table

Document Number / Title	Revision / Release Date	Description
ESP32 TRM (Version 5.5)	2025.08	<i>ESP32 - Technical Reference Manual Version 5.5</i>
ISO 9241-210	2019	<i>Ergonomics of human-system interaction</i>
ISO/IEC 14882	2024	<i>Programming languages — C++</i>
IEEE 802.11	2020	<i>IEEE Standard for Information Technology--Telecommunications and Information Exchange between Systems</i>
ISO/IEC 9075-1	2023	<i>Information technology — Database languages SQL</i>

2.3. Order of Precedence

In the event of a conflict between the text of this specification and an applicable document cited herein, the text of this specification takes precedence without any exceptions.

All specifications, standards, exhibits, drawings or other documents that are invoked as “applicable” in this specification are incorporated as cited. All documents that are referred to within an applicable report are considered to be for guidance and information only, except ICDs that have their relevant documents considered to be incorporated as cited.

3. Requirements

3.1. System Definition

The Smart Solar Residential Lighting and Security System integrates four main subsystems, each fulfilling a critical role in delivering a reliable, energy-efficient solution for residential applications. The solar power network centers on an MPPT charge controller that optimizes energy harvesting by regulating the variable DC output from the solar panels to efficiently charge the battery. This controller communicates with the battery management systems and central microcontroller to ensure continuous power availability.

The physical control subsystems provide a user-friendly touch screen interface for lighting control and essential system status monitoring. It also hosts DC voltage conversions and serves as the central connection point for all system components, ensuring efficient power distribution and safe operation compliant with residential electric codes.

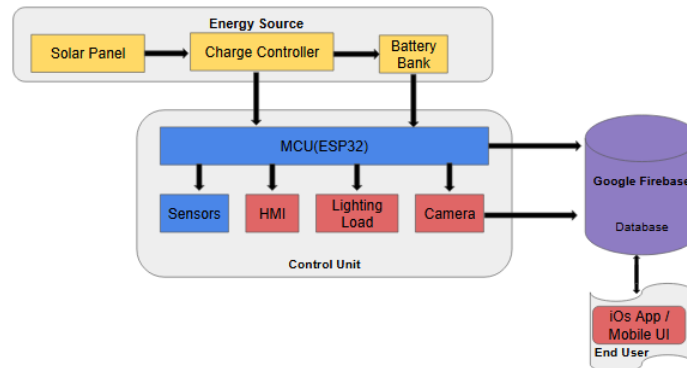
The lighting and security monitoring functions are supported by an android application paired with an ESP32-EYE camera module. This subsystem streams live video feeds and

uses AI-powered motion detection to alert users of security events, while securely logging data to a cloud database for later review.

At the core of the system, the microcontroller and sensor subsystem collect environmental and motion data, execute control logic, and maintain real-time communication with the mobile app and cloud database via Wi-Fi. Sensor inputs trigger lighting adjustment and security alerts, ensuring a responsive and adaptive system behavior.

Together, these interconnected subsystems create a cohesive, scalable solution that provides solar-powered, automated lighting and security, addressing the need for sustainable energy use and enhancing residential awareness.

Figure 3.1.1. Block Diagram of System



3.2. Characteristics

3.2.1. Functional / Performance Requirements

3.2.1.1. Lighting Duration

The system shall provide continuous home/residential operation of exterior and interior lighting for a minimum of eight hours during night house conditions, powered solely from solar energy.

Rationale: This ensures that residential homes can receive reliable illumination throughout the night that is not dependent on grid electricity. This is critical to both the usability of the system and ability to reduce monthly costs.

3.2.1.2. Motion Detection Accuracy

The system will be able to detect and log human motion events with an accuracy of at least 90% under typical residential operating conditions, including varying lighting level and regular environmental noise

Rationale: Reliable detection of motion is essential for the systems role as a resource for security. A reliable, High detection accuracy will minimize false negatives and ensures timely push notifications that are generated for the homeowner

3.2.1.3. Camera Streaming Latency

The mobile android application developed in Kotlin shall stream live video from the ESP32-EYE camera with no more than two seconds when operating over standard WiFi

Rationale: The requirement will ensure that the homeowner experiences video feedback that is as close to real time as possible for responsive decision making for the user

3.2.1.4. Database Storage

The system shall maintain at least thirty days of stored event snapshot data in its cloud (firebase) for later review and analysis

Rationale: A month of storage strikes between useful historical/security data and practical storage constraints, allowing homeowners to review the past without over the top resource costs.

3.2.1.5. User Control and Responsiveness

The Android application will allow the user to toggle lighting states (ON / OFF) and received push notifications of change within 2 to 3 seconds of command execution

Rationale: Ensures the mobile will feel seamless and responsive to the user, which is essential for modern usability

3.2.1.6. Motion Triggered Lighting

The system shall activate exterior lighting once triggered by motion detection for at least 30 seconds unless reset by further activity

Rationale: Automatic light response provides both security and deterrence. This will provide users more safety and controllability within the application.

3.2.1.7. System Start-Up Sequence

The system shall complete its full start-up sequence within 10 seconds from the application of power, including initialization of the microcontroller, sensors, wireless communications, and user interface.

Rationale: Defines user expectations for system responsiveness and ensures the system becomes operational in a predictable amount of time. The PIR sensor shall stabilize within 30 seconds of power-on to allow accurate motion detection. PIR sensors require a warm-up period to avoid false triggers. The camera module shall begin streaming and AI-based recognition within 3 seconds after the system boots. Allows camera-based alerts to function shortly after startup without delaying system security.

3.2.1.8. System Start-Up Sequence

The system shall enter a safe shutdown mode within 5 seconds of power removal or command shutdown, ensuring lighting loads are disabled and data logging is halted without corruption.

Rationale: Prevents unintended lighting activation and protects data integrity under unexpected power loss.

3.2.1.9. Hardware and Mobile Interface Interaction

The system shall maintain a bidirectional communication interface between the hardware control units (MCU, sensors, lighting modules, and ESP-32 camera) and the android application. All hardware status signals should be updated and transmitted to the application in real time with a less than 5 seconds delay.

Rationale: This requirement clarifies the necessary interaction pathways between physical hardware components and the mobile application, ensuring that users receive accurate system status and can remotely control all major functions in a timely manner.

3.2.2. Physical Characteristics

3.2.2.1. Subsystem Size

The physical control unit shall not exceed 15 x 20 cm footprint, while individual sensors and camera shall fit within a 10x10x10 cm volume.

Rationale: Ensures compact sizing for ease of installation in tight residential spaces while maintaining discreet placement for cameras and sensors

3.2.2.2. Battery Housing

The battery housing shall be fully enclosed in a weather resistant casing that is also ventilated

Rationale: Protects deep cycle battery from weather exposure while allowing airflow

3.2.2.3. Cable Management and Mounting

All interconnections between the subsystems shall utilize standard cable runs and shall be designed for mounting with basic tools

Rationale: Simplifies installation and ensures compliance with safe residential practices

3.2.3. Electrical Characteristics

3.2.3.1. Inputs

- a. The Solar Home Lighting System shall tolerate any combination and sequence of input signals as defined in the Interface Control Document (ICD) without sustaining damage, degradation of life expectancy, or operational malfunction. This resilience applies equally whether the system is energized or de-energized.

- b. No permissible sequence of input commands shall induce system failure, reduce expected service life, or impair functionality.

Rationale: The system is engineered to minimize the risk of damage or malfunction arising from user or technician actions, ensuring reliability in real-world operation.

3.2.3.1.1. Table of Inputs

Parameter	Requirement	Notes
Input Voltage Range	12–24 VDC	Compatible with solar batteries and MPPT output
System Peak Power	≤ 50 W	From requirement 3.2.3.1.1
Converter Efficiency	$\geq 85\%$	Applies to all DC/DC converters
Input Ripple Tolerance	0–200 mV	Based on converter datasheet limits
Output Regulation (Logic Rail)	$3.3\text{ V} \pm 3\%$	ESP32 logic supply
Lighting Output Regulation	12 VDC	Supports LED loads
Overcurrent Protection	Required	Per requirement 3.2.3.1.4
Short-Circuit Protection	Required	Must survive without damage
Reverse Polarity Protection	Required	Diode or active method
Thermal Derating	Components shall operate within -20°C to $+55^{\circ}\text{C}$	Matches environmental requirement

3.2.3.1.1. Power Consumption

The maximum peak power of the system, including operation of lights, camera, sensors, wireless communication, shall not exceed 50 watts

Rationale: Limiting peak power draw ensures easy compatibility with mid level solar generation capacity and affordable battery storage. This will enable the system to remain cost effective while still meeting performance requirements

3.2.3.1.2. Input Voltage Range

The system shall accept DC input from 12 to 24 volts, compatible with standard solar batteries

Rationale: Flexibility will allow for seamless integration with commercially available solar panels and storage systems without custom hardware

3.2.3.1.3 DC/DC Conversion Efficiency

All DC/DC converters (bucks and boost) shall maintain a minimum conversion efficiency of 85% under load conditions

Rationale: High efficiency is required to minimize energy loss and maximize usable solar energy

3.2.3.1.4. Overcurrent and Short Circuit Protection

The system shall include overcurrent protection on all major supply rails and shall withstand short circuit episodes without permanent damage to the system

Rationale: Protects the system from wiring faults, load failures, and user error. This will ensure safety and reliability

3.2.3.1.4. Data and Control Signals

All MCU and sensor interfaces should conform to logic levels of 3.3 V with proper protection for I/O pins

Rationale: Aligns with ESP32 electrical limits, ensuring proper and safe communication between subsystems

3.2.3.2. Outputs

3.2.3.2.1. Energy Output

The Solar Home Lighting System shall provide a regulated DC output compatible with 12V appliances.

Rationale: Ensures LED lighting can be used off-grid.

3.2.3.2.2. Battery Status Output

The Solar Home Lighting System shall include a digital interface to provide the battery charge level.

Rationale: The user and technician can monitor battery performance and plan accordingly.

3.2.3.2.3. Raw Video Output

The Solar Home Lighting System application shall include a raw video interface to support external recording.

Rationale: Too much data to store videos; only photos are stored internally. It would be used for security.

3.2.3.2.4. Connectors

The Solar Home Lighting System shall use weatherproof DC connectors compliant with IEC 62852.

Rationale: Ensures safe connections in outdoor environments.

3.2.3.2.5. Wiring

The Search and Rescue System shall follow the guidelines outlined in NEC Article 690 and use UL-listed cables.

Rationale: Ensures electrical safety and minimizes fire risk.

3.2.4. Environmental Requirements

The Solar Home Lighting System shall be designed to withstand and operate in the environmental conditions and laboratory tests specified in the following subsections.

Rationale: These requirements are based on scenarios where the system must maintain reliable operation under varying and harsh weather.

3.2.4.1. Thermal

The Solar Home Lighting System shall operate between -20°C to $+55^{\circ}\text{C}$.

Rationale: This range accommodates extreme hot and cold conditions in the U.S.

3.2.4.2. Rain

The Solar Home Lighting System shall withstand heavy rainfall and storm conditions, including exposure to high precipitation rates.

Rationale: Ensures waterproofing and short circuits, and corrosion.

3.2.4.3. Solar Radiation

All components exposed to outdoor elements shall be UV-resistant and capable of withstanding high solar irradiance.

Rationale: Prolonged exposure to sunlight can degrade plastics and coatings of components.

3.2.4.2. Wind

The Solar Home Lighting System shall withstand 150 km/h wind speeds without structural failure and detachment.

Rationale: High winds pose a risk of damage to rooftop installations.

3.2.5. Failure Propagation

The Solar Home Lighting System shall not allow propagation of faults beyond its system interface, ensuring that failures within the system do not affect connected external devices or the grid interface.

3.2.5.1. Failure Detection, Isolation, and Recovery (FDIR)

3.2.5.1.1 Built-In Test (BIT)

The Solar Home Lighting System shall include an internal subsystem that generates diagnostic signals and evaluates system responses to determine the presence of faults.

3.2.5.1.1.1. BIT Critical Fault Detection

The BIT shall detect critical faults in the Solar Home Lighting System with a minimum detection rate of 95 percent.

Rationale: This requirement ensures reliable fault detection in remote or off-grid environments where manual diagnostics are impractical.

3.2.5.1.1.2. BIT False Alarms

The BIT shall have a false alarm rate of less than 5 percent.

Rationale: Minimizing false alarms reduces unnecessary maintenance interventions and improves user trust in system diagnostics.

3.2.5.1.1.3. BIT Log

The BIT shall record the results of each diagnostic test in a log stored within the Solar Home Lighting System. This log shall be accessible for retrieval and clearing by maintenance personnel.

Rationale: Persistent logging supports fault analysis and long-term reliability tracking.

3.2.5.1.2 Isolation and Recovery

The Solar Home Lighting System shall support fault isolation and recovery by enabling subsystems to be reset, disabled, or reconfigured based on BIT results.

Rationale: This capability ensures that faults can be contained and mitigated without requiring full system shutdown, preserving partial functionality in critical lighting scenarios.

4. Support Requirements

The Smart Solar Residential Lighting and Security System requires stable Wi-Fi connection within the home to enable communication between the control panel and mobile applications. Users must provide access to Wi-Fi and maintain the solar panel installation in an area with sufficient sunlight for optimal performance. A compatible Android device is additionally required to operate the mobile application and receive security notifications.

The system package includes all necessary hardware components, such as solar panels, batteries, control boards, sensors, and lighting units. Users will use the Android mobile application for wireless control and monitoring. A comprehensive user manual and an in-app walkthrough guide are provided to assist with installation, operation, and troubleshooting.

Technical support is available via email and phone during standard business hours. In the event of hardware malfunction or operational issues in the field, users can contact support for remote diagnostic and troubleshooting. Field issues related to solar panel position or battery performance may require an on-site inspection or user intervention as guided by support. Periodic software updates will be pushed automatically to enhance system functionality and security.

Appendix A: Acronyms and Abbreviations

MPPT	Maximum Power Point Tracking
PWM	Pulse Width Modulation
Wi-Fi	Wireless Fidelity
AC	Alternating Current
DC	Direct Current
RMS	Root Mean Square
HMI	Human Machine Interface
ICD	Interface Control Document
MCU	Microcontroller Unit
km/h	Kilowatts per hour
LED	Light Emitting Diode

Appendix B: Definition of Terms

This appendix provides definitions for specialized technical terms and acronyms used throughout the Functional System Requirements document. These definitions ensure consistency and clarity when interpreting the system description, requirements, and validation criteria.

B. Definitions of Terms

Solar Irradiance	The amount of solar power received per unit area, measured in watts per square meter (W/m^2).
MPPT (Maximum Power Point Tracking)	A control method used by the charge controller to continuously adjust voltage and current for maximum solar-panel power output under varying sunlight conditions.
Lux	A unit of illuminance equal to one lumen per square meter, used to quantify visible light intensity detected by the system's ambient-light sensor.
HMI (Human-Machine Interface)	The touchscreen display and control interface that allows users to view system data and issue lighting or configuration commands.
Firebase	A cloud-based database service used to store sensor data, camera images, and system status information for mobile-app access.
ESP32	A low-power microcontroller with integrated Wi-Fi and Bluetooth, used as the system's primary processing and communication unit.